

Research Methodology



2015

2015

Differentiate b/w bibliography and
reference?

Book pg → 204

2015

Sampling frame: and its effect
on survey

book pg → 144

Effect on survey-

- Representation
- Bias and error
- Data Quality
- Efficiency

~ iii ~

Write a note on cluster Sampling?

Book pg → 135, 136

~ iv ~

Define pretesting of a questionnaire
What are the advantages?

Book pg → 126

~ Long Question ~

~ QN02 ~

Explain the term 'Survey' and its various types?

Book pg → 123

~ QN04 ~

How Sample Size is determined under different Condition
Simpl size → Book pg 145 (Small paragraph)

(1) Desired Confidence Level

- Higher Confidence Level:- Requires a larger Sample size to ensure the results are more reliable.
- Common Confidence Level:- 90%, 95% and 99%.

(2) Margin of Error:-

- Smaller margin of error:- Requires a larger sample size to achieve more precise estimates
- Common margins of error:- 1%, 5% and 10%

(3) population variability:-

- Higher variability:- Requires a larger sample size to capture the diversity within the pop
- Lower variability:- smaller sample size may be sufficient

(4) population size:-

- Large population:- sample size may not need to be proportionally large
- Small population:- small size may need to be a significant proportion of the population

(5) Effect Size:-

- Larger Effect size:- smaller sample size may be sufficient
- Smaller effect size:- larger sample size is needed to detect

Q No 3

What are the qualities of good data? Explain its type and collect.

Solution:-

Qualities of good data:-

- Accurate:- The data should be correct and free from errors.
- Complete:- All necessary data should be included.
- Consistent:- Data should be in the same format and stable over time.
- timely:- Data should be up-to-date and collected at the right time.
- Relevant:- The data should be related to the topic or problem being analyzed.

Types of data Collection:-

- Survey and Questionnaires:- you ask people questions and they give their answers.
- Interview:- you talk to people directly and get detailed responses.

- Observations:- you watch and record what people do
- Experiment:- you perform test and record the results
- Document Review:- You look at existing document and gather information
- Focus Group:- you have discussions with a small group of people to get their opinions

Sources:-

Internal Sources:- Book pg (101)

External Sources:- Book pg (101)

Types:-

Primary Data:- on pg (103) Book

Example:- on previous page

(Type of data collection)

Secondary Data:-

on pg 101 (Book)

(2016)

(1) Nominal and ordinal Scale:-

Book

pg → 82 and 83

(2) Semantic differential and Likert Scale

Book pg → 93 and 95

(3) Cross sectional and longitudinal study

pg → 13, 14

(4) Close ended questions:-

Book pg → (113)

Long Question:-

Q/2 Define and explain under what kind of Condition they are used

(a) stratified sampling → pg 134

(b) systematic sampling → pg 137

(c) SRS → pg 134

Q/3 Explain the errors that affect accuracy. Also, give the reasons how they can be avoided

pg (86 → 89)

(2017)

(i)

Sampling frame:-
Repeat → 2015

(ii)

Census:-

pg → 125

(iii)

presenting for Questionnaire:-
Repeat → 2015

(iv)

Literature review:-

Book pg → 178

Long Question

Q/2:- Explain open and close ended questions. Their advantage and disadvantage and the situation in which they are used?

Book Pg → 112

Q No 3:- Repeat 2015 → (Q/4)

~ (2018) ~

(a) stratified Sampling.- Repeat

(b) Measurement error-

Pg → (88)

~ (c) ~

processing of Survey Data:-

- Collect response:- Gather answers from your survey.
- Enter Data:- put the collat answers into a digital format like a spreadsheet
- Clean Data:- check for any mistake or missing information and fix them
- Code Data:- Convert text answer into numbers, if needed, to make Analysis easier
- Transform Data:- make any necessary adjustment like calculating average
- store Data:- save your cleaned and coded data securely

Types of data Collection:-

Repeat → 2015

Long Question

QNo2 Pg (110)

What is a questionnaire? Which factors should be considered in designing of a questionnaire? Explain in detail.

Q/3 Pg (82)

Define Scale and its types. Also explain different types of rating Scales by giving at least two example each.

PUACP

(2019)

(a)

Likert Scale:-

Repeat

Itemized Ratio Scale

An itemized rating scale is a type of closed-ended question that provides respondents with a set of fixed options to choose from each representing different levels of a particular attribute. These scales help quantify subjective information, making it easier to analyze and interpret.

(b)

Systematic Sampling:-

Repeat

(c)

Coverage Error:-

Coverage Error occurs when some members of the

population are not included in the sampling frame, leading to biased survey results.

Method to collect Data

- 1 Survey and Questionnaires
- 2 Interview
- 3 Observations
- 4 Experiment

Source measurement Error:-

Repeat pg (88)

~~add~~

Accuracy in Survey:-

Accuracy in Survey refers to the degree to which the collected data correctly represents the true characteristics, opinion or behaviours of the population being studied.

- (i) Question Design
- (ii) Sampling Method
- (iii) Data Collection

Long Question:-

QNo 3:-

Explain ten steps for
a successful survey?

Book pg (126)

QNo 4:-

Write in detail about the
essential components of good research
proposal and what are the factors
affecting the research design?

research proposal → 177

research Design → 60

QNo 5:-

What are various types of
questions asked while conducting a
research? Discuss their advantages
and disadvantages.

pg 112 & 113

2020

2020

Steps to Conduct a Successful Survey? pg → 126

2020

Marketing Survey:-

Conducting a marketing survey is a great way to gather valuable insights about your target audience, their preferences, and behaviors.

Steps to Conduct a Marketing Survey:-

- 1) Define Objectives
- 2) Identify Target Audience
- 3) Design the Survey
- 4) Choose a Survey Method
- 5) Distribute the Survey
- 6) Collect And Analyze Data
- 7) Report findings.

(c)
Methods of sample selection
pg → 134, 135, 137, 138, 139

(d)
Scaling techniques:-
pg → 82, 83, 84, 93, 95

Long Question:-

QNO3:-

Describe different types and methods of Conducting research?

Pg 3, 11, 12, 15 → research

pg 7, 8 → method

QNO4:-

Describe the Importance of sample size in a research and also discuss various methods of sample size determination?

Repeat → 2015, 2017

QNO5:-

Discuss the styles and formats of a research Report?

Book pg (192-197)

~(2021)~

~(QNO1)~

~(i)~

Sampling frame and its requirements.
Repeat → 2015, 2017

~(ii)~

Criteria of good research:-

Good research is characterized by several key criteria that ensure its quality, reliability and validity.

- 1) Clear objectives
- 2) Relevance
- 3) validity
- 4) Reliability

~(iii)~

Test of validity for a measurement?

- 1) face validity
- 2) Content validity
- 3) Construct validity
- 4) Criterion validity
- 5) Internal validity
- 6) External validity

(iv)

Difference between Questionnaires and Schedule?

Questionnaires

- Respondents Answer Questions on their own

- questions are presented in a written or online format

- questions are pre-defined and follow a specific order

Schedule

- A trained interviewer Asks Questions and records responses.

- Interviews are conducted in person or over the phone

- Question may be predefined, but the interviewer may also ask follow-up questions

~(v)~

Significance of report writing:-
Repeat

~(vi)~

Open Ended Questions
Repeat

~(Long Question)~

~(QNO2)~

Explain the meaning of research
also describe the various type
of research?

pg. 60 to 63

~(QNO3)~

What do you meant by Research
Design? What point should be taken into
Consideration by a researcher in developing
a sample design for a research project?

Research Design:-

Research design is a framework or blueprint for conducting a research project. It outlines the procedures for collecting, analyzing, and interpreting data.

point to consider in developing a Sample Design:-

- (1) Define the population:- Clearly identify the from which the sample will be drawn.
- (2) Sampling frame:- Ensure the sampling frame is comprehensive and up-to-date to avoid coverage error.
- (3) Sample size:- Determine an appropriate sample size based on the research objectives, desired confidence level, and margin of error.
- (4) Sampling method:- Choose a suitable sampling method to ensure representativeness.
- (5) Data collection method:- Decide on the method of data collection that best suits the research objectives.

(6) Ethical Consideration:- Ensure

Ethical standards are maintained

(7) Budget and Resources:- Consider the available budget and resource for conducting the research.

QNO 4

What are the techniques for interpretation of research results?

Describe the precautions that researcher should take while interpreting his findings?

Solution:-

→ Techniques for interpretation of Research Results:-

(1) Descriptive Statistics:-

Use measures like mean, median, mode, Standard deviation and range to summarize and describe data.

(2) Inferential statistics:-

Employ statistical test (eg t-test, F-test, chi-square test, Anova) to make inference about the population based on sample data

(3) Correlation Analysis:- Examine relationship between variable using Correlation coefficients to understand the strength and direction of associations.

(4) Content Analysis:- for qualitative data, Analyze patterns, themes, and categories to interpret the underlying meanings and variables.

(5) Comparative Analysis:- Compare the results with previous studies to identify similarities, differences, and trends.

→ Precautions to take while interpreting Findings:-

(1) Consider Limitations:- Acknowledge the limitations of the study include Sample Size methodology and potential Confounding Variables.

(2) Contextual Relevance:- Interpret the results in the context of the research question and the broader field of study.

(3) Use Appropriate statistical Test:- Ensure that the correct statistical test are used based on the nature of the data and the research design.

(4) Cross-verification:- validate findings through peer review, replication studies, or by consulting with experts in the field.

2022

Explain the Sources of Secondary data?

Book pg → 101

Telephonic Survey:-

A telephonic survey is a method of a data collection when researchers conduct interviews

over the phone to gather information from respondents. The approach is widely used in market research, opinion polling, and Social Science studies.

Advantage:-

- (1) Cost effective
- (2) Time effective
- (3) Geographical results
- (4) Flexibility

DisAdvantage:-

- (1) Non-response
- (2) Sampling bias
- (3) technical issues:-

Explain Discriminant Analysis:-

Discriminant analysis is a statistical technique used to classify observations into predefined groups based on their characteristics. It is commonly used in situations where the dependent variable is categorical and the independent variables are continuous or categorical.

~Q4~

D/w schedule and questionnaire?
Repeat

~Q5~

Different types of rating scale?
pg 93, 95, 96

~Q6~

What is difference b/w reference
and bibliography?

Repeat Book pg → 204

Long Question

~QNO2~

Explain the type of error in
Sample Survey. Describe the different
methods to control each type
of error?

from Book pg (86 to 89)
all error

QNO3

What is literature review? Write its important and function. Explain the procedures for reviewing the literature.

Ans:-

Literature Review:-

A literature review is a Comprehensive Summary and Analysis of the existing research on a particular topic. It involves identifying, evaluating.

→ Importance of Literature Review:-

- (1) Identification of Gaps:- Highlights gaps in the current research, identifying areas that need further investigation.
- (2) Avoiding of Gaps:- Ensure that the research does not duplicate existing that studies, saving time and resource.
- (3) Theoretical framework:- Helps in developing a theoretical framework for the study by identifying key concepts and theories.

→ Functions of Literature Review:-

- (1) Summarization:- Summarizes the key finding of previous research.
- (2) Synthesis:- Combines the finding of different studies to provide a comprehensive understanding.
- (3) Identification of trend:- Identifies trends and patterns in the literature.

→ Procedure for Reviewing the Literature:-

- (1) Define the scope:- Clearly define the scope and objectives of the literature review.
- (2) Search for literature:- Conduct a comprehensive search for relevant literature using database, books and other sources.
- (3) Selective relevant studies:- Screen the identified literature to select studies that are relevant to the research topic.

Q No 4

What is pretesting of questionnaires?
write Adv and dis Adv of
pretesting of questionnaires?

Book pg (126)
(110 — 113)

2023

i)

Advantages and disadvantages of
mail surveys?

Book pg (108 — 109)

ii)

Stratified random sample:-

Repeat

PUACP

iii)

Describe the precautions that
researcher should take while
interpreting his findings?

Avoid over-interpreting results and
check for bias. Ensure data quality
and know your research design's limitations.

Don't over-generalize and think
about alternative explanations. Be
aware of culture and social biases
report limitations, provide clear results
and disclose conflicts of interest.

Stay objective, document your process,
and seek peer review.

(iv)

Explain bibliography and write its
importance in context of research
report?

Book pg → 203

(v)

What are the characteristics of research
report?

(vi)

Explain quantitative research and
qualitative research with the help
of examples?

QNo2

What do you mean by research design?
Differentiate between research methodology
and research design. illustrate your
answer with suitable example?

pg 55, 53, 2

QNo3

Discuss the qualities of well
designed questionnaire. Describe
the various steps which are
used in designing a questionnaire

pg → 110

QNo4

Explain the essential components
of good research proposal?
What are the major sections
of writing a research proposal?

pg → 177